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ScienceDirect
Journal of Ethnopharmacology 111 (2007) 197–201
Journal of ETHNO-PHARMACOLOGY
www.elsevier.com/locate/jethpharm

Activity of essential oils from Brazilian medicinal plants on *Escherichia coli*

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Received 28 January 2005; received in revised form 23 October 2006; accepted 6 November 2006
Available online 3 December 2006

Abstract

Essential oils obtained from leaves of 29 medicinal plants commonly used in Brazil were screened against 13 different *Escherichia coli* serotypes. The oils were obtained by water-distillation using a Clevenger-type system and their minimal inhibitory concentration (MIC) were determined by microdilution method. Essential oil from *Cymbopogon martinii* exhibited a broad inhibition spectrum, presenting strong activity (MIC between 100 and 500 μg/mL) against 10 out of 13 *Escherichia coli* serotypes: three enterotoxigenic, two enteropathogenic, three enteroinvasive and two shiga-toxin producers. *C. winterianus* inhibited strongly two enterotoxigenic, one enteropathogenic, one enteroinvasive and one shiga-toxin producer serotypes. *Aloysia triphylla* also shows good potential to kill *Escherichia coli* with moderate to strong inhibition. Other essential oils showed antimicrobial properties, however with a more restricted action against the serotypes studied. Chemical analysis of *Cymbopogon martinii* essential oil performed by gas chromatography (GC) and gas chromatography/mass spectrometry (GC–MS) showed the presence of compounds with known antimicrobial activity, including geraniol, geranyl acetate and *trans*-cariophyllene, which tested separately, indicated geraniol as antimicrobial active compound. The significant antibacterial activity of *Cymbopogon martinii* oil suggests that they could serve as a source for compounds with therapeutic potential.

Keywords: *Escherichia coli*; *Cymbopogon martinii*; *C. winterianus*; *Aloysia triphylla*; Essential oil; Antimicrobial activity

1. Introduction

Diarrhoea caused by *Escherichia coli* infection is an emergent problem in both developing and developed world and is responsible for high rates of mortality in newborn children and animals (Radu et al., 2001). Although commensal representatives found in the intestinal flora of man and animals are non-pathogenic, certain strains are highly pathogenic.

All diarrhoeagenic strains of *Escherichia coli* were initially termed enteropathogenic *Escherichia coli* (EPEC) but as their pathogenic mechanisms were known, they were grouped into seven classes (Clarke, 2001). There are now seven classes, namely enteropathogenic *Escherichia coli* (EPEC), enterohaemorrhagic *Escherichia coli* (EHEC), enteroinvasive *Escherichia coli* (EIEC), enterotoxigenic *Escherichia coli* (ETEC), enteroaggregative *Escherichia coli* (EaggEC), diarrhoea-associated hemolytic *Escherichia coli* (DHEC) and cytolethal distending toxin (CDT)-producing *Escherichia coli* (Kaper, 1994).

According to Schmidt et al. (1997), EPEC and ETEC are the most important of these in terms of total diarrhea episodes on a global scale although EHEC has become more significant in developed countries in recent years due to the occurrence of numerous outbreaks. Strictly, the term “EPEC” should be reserved for those *Escherichia coli* which do not belong to any of the other diarrhoeagenic *Escherichia coli* classes and do not produce any enterotoxin but do cause characteristic attaching and effacing lesions on the intestinal brush border (Edelman and Levine, 1983).

ETEC are nowadays considered a major cause of *Escherichia coli*-associated diarrhoea worldwide, producing one or both of two types of enterotoxins namely heat stable enterotoxins (ST) and heat labile enterotoxins (LT) (Clarke, 2001).

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doi:10.1016/j.jep.2006.11.034

Table 1
Identification, voucher specimen and data on traditional use of the plants studied

Botanical name	Family	Voucher	Origin[a]	Plant part[b]	Preparation/traditional use[c,d]
Achyrocline satureioides (Lam.) DC.	Asteraceae	UEC 127.116	N	lv	I: infusion/used as anti-inflammatory and anti-infective (antimicrobial and antiviral)
Allium schoenoprasum L.	Liliaceae	UEC 121.397	E	lv	I: fresh juice/antiseptic properties; used to promote the digestive processes
A. tripilla (L'Hér.) Britton	Verbenaceae	UEC 121.412	E	lv	E: tincture or essential oil/used as bactericide; applied to skin treatment
Anthemis nobilis L.	Asteraceae	UEC 121.411	E	lv	I: infusion/used as antispasmodic and digestive; also as food preservative
Artemisia annua L.	Asteraceae	CPQBA 1246	E	lv	E: tincture/applied as disinfectant and antiseptic; I: extract/antimalaric
Artemísia camphorata L.	Asteraceae	CPQBA 63	E	lv	I: Infusion/used as antiseptic and sedative
Baccharis trimera DC.	Asteraceae	CPQBA 1	N	lv	I: infusion/treat digestive tract disorders and viral infections (herpes simplex)
Cordia verbenacea DC.	Boraginaceae	UEC112744	N	lv	E: tincture/effective as anti-inflammatory
Cymbopogon martinii (*Roxb.*) *W. Watson*	Poaceae	UEC 127.115	E	lv	E: tincture or essential oil/used as antiseptic and insect repellent
C. winterianus Jowitt	Poaceae	UEC 121.414	E	lv	E: tincture or essential oil/used as antiseptic and insect repellent
Cyperus articulatus L.	Cyperaceae	UEC 121.396	N	rt	E: decoction/used as anti-infective and anti-inflammatory
C. rotundus L.	Cyperaceae	CPQBA 1252	N	rt	E: decoction/used as anti-infective and anti-inflammatory
Lippia alba (Mill.) N. E. Br. ex Britton & P. Wilson	Verbenaceae	UEC121413	N	lv	I: infusion/employed as soothing and for headaches
M. piperita L.	Lamiaceae	UEC 127.110	E	lv	E: fresh juice/antiseptic and anti-infective; I: vermifug
Mikania glomerata Spreng.	Asteraceae	UEC 102047	N	lv	I: syrup/used as anti-infective, anti-inflammatory and bronchodilator
M. laevigata Sch. Bip. ex Baker	Asteraceae	UEC 102044	N	lv	I: syrup/used as anti-infective, anti-inflammatory and bronchodilator
Ocimum basilicum L.	Lamiaceae	UEC 121.408	E	lv	I: infusion/usefulness associated with digestive tract disorders
O. gratissimum L.	Lamiaceae	UEC 121.407	N	lv	I: infusion/used for colds and as diuretic
Origanum × *applii* Boros	Lamiaceae	UEC 121.410	E	lv	I: infusion/analgesic and expectorant
O. vulgare subsp. *virens* L.	Lamiaceae	UEC 121.409	E	lv	I: infusion/analgesic and expectorant
Piper abutilodes Kunth	Piperaceae	UEC 127.122	N	lv	I: tinture/hepatic an digestive tract disorders
P. aduncum L.	Piperaceae	UEC 127.118	N	lv	I: tincture/astringent and used for digestive tract disorders
P. marginatum Jacq.	Piperaceae	UEC 121.395	N	lv	I: tincture/astringent and used for digestive tract disorders
P. molicomum Kunth	Piperaceae	UEC 127.121	N	lv	I: tinture/hepatic and used for indigestion
P. regnellii (Miq.) C. DC.	Piperaceae	UEC 127.120	N	lv	I: tincture/astringent and used for digestive tract disorders
Stachys byzantina K. Koch	Lamiaceae	UEC 121.404	E	lv	E: infusion/anti-inflammatory
Stachytarpheta cayennensis (*Rich.*) *Vahl*	Verbenaceae	UEC 121.394	N	lv	I: infusion/used as anti-inflammatory, antihelmintic and antiulcerogenic
Thymus vulgaris L.	Lamiaceae	UEC 121.405	E	lv	I: infusion/antiseptic and for digestive tract disorders
Urena lobata L.	Malvaceae	CPQBA 1251	E	lv	I: Infusion/mouth antiseptic and sore throat

[a] N, native to Brazil; E, exotic.
[b] lv, leaves; rt, roots.
[c] Data from Lorenzi and Matos (2002).
[d] http://www.rain-tree.com/; I: internal use; E: external use.

The diarrhoea is often treated with antibacterial drugs, but this treatment is generally ineffective, due in part to the presence of drug resistant strains and failure to identify drug sensitivity (Cid et al., 1996). This fact has underscored the need of a quick development of antibacterial drugs that are more effective than those currently in use.

Since medicinal plants play a fundamental role in traditional medicine, the use of herbal remedies has been increased in Brazil as well as in another countries. However, according to Cowan (1999), while 25–50% of current pharmaceuticals are derived from plants, none are used as antimicrobials. Thus, screening programs for antibacterial activity may yield candidate compounds for developing new antimicrobial drugs (Ahmad and Beg, 2001).

In the present study, essential oils from 29 native and exotic medicinal plants traditionally used in Brazil, most of them with folk indication as anti-infective or anti-septic were screened for antimicrobial activity against different *Escherichia coli* serotypes. The essential oil from *Cymbopogon martinii* was also characterized by gas-chromatography/mass spectrometry analyses in order to identify the major compounds.

2. Materials and methods

2.1. Plant material

Table 1 shows the plants studied, including the botanical name, voucher specimen and data related to traditional use and plant parts used. The plants were grown in the experimental field of the Research Center for Chemistry, Biology and Agriculture (CPQBA, State University of Campinas, São Paulo, Brazil). The plants were collected from November 2001 to February 2003. Voucher specimens were deposited at the State University of Campinas Herbarium (UEC), and identified by Dr. Washington Marcondes Ferreira Neto (curator), or at the herbarium at the CPQBA.

2.2. Essential oil extraction

The essential oils were obtained from 40 g of fresh plant parts by water-distillation using a Clevenger-type system for 3 h. The aqueous phase was extracted three times with 50 mL dichloromethane. The pooled organic phases were dried with sodium sulphate, filtered and the solvent evaporated until dryness. Dried samples were stored at −25 °C in sealed glass vials. Prior GC–MS analysis the samples were solubilized in ethyl acetate (10 mg/mL).

2.3. Antibacterial assay: minimal inhibitory concentration (MIC) test

Escherichia coli serotypes were kindly supplied by Dr. Tânia A. Tardelli Gomes do Amaral from Biomedical Sciences Department, Federal University from São Paulo, Brazil. The bacteria were grown overnight at 36 °C in McConkey Agar (Merck), and inoculum for the assays was prepared by diluting cell mass in 0.85% NaCl solution, adjusted to McFarland scale 0.5 and confirmed by spectrophotometrical reading at 580 nm. Cell suspensions were finally diluted to 10^4 UFC mL^{-1} for being used in the activity assays.

MIC tests were carried out according to Ellof (1998), using Nutrient Broth or Müller-Hinton Broth on a tissue culture test-plate (96 wells). The media was chosen according to serotypes growth response. The stock solutions of the extracts and oils were diluted and transferred into the first well, and serial dilutions were performed so that concentrations in the range of 1000–16.0 μg/mL were obtained. Chloramphenicol (Merck) was used as the reference antimycotic control in the range of 62.5–0.5 μg/mL. The inoculum was added to all wells and the plates were incubated at 36 °C for 48 h. Antimicrobial activity was detected by adding 20 μL of 0.5% TTC (triphenyl tetrazolium chloride, Merck) aqueous solution. MIC was defined as the lowest concentration of oil that inhibited visible growth, as indicated by the TCC staining (dead cells are not stained by TTC).

2.4. Gas chromatographic (GC) and mass spectrometry (GC–MS) analyses

The identification of volatile constituents was performed using a Hewlett-Packard 5890 Series II gas chromatograph, equipped with a HP-5971 mass selective detector and capillary column HP-5 (25 m × 0.2 mm × 0.33 μm diameter). GC and GC–MS were done using *split/splitless* injection, with injector set at 220 °C, column set at 60 °C, with heating ramp of 3 °C $min^{-1}$and final temperature 240 °C for 7 min, and the FID detector set at 250 °C. Helium was used as carrier gas at 1 mL min^{-1}. The GC–MS electron ionization system was set at 70 eV. A sample of the essential oil was solubilized in ethyl acetate for the analyses. Retention indices (RI) were determined by co-injection of hydrocarbon standards. The oil components were identified by comparison with data from literature (Adams, 2001), the profiles from the Wiley 138 and Nist 98 libraries, and by co-injection of authentic standards, when available.

3. Results and discussion

3.1. Oil yields

Oil yields of the plant parts indicated in Table 1, expressed in relation to dry weight plant material, are presented in Table 2. Larger quantities were obtained from *Aloysia triphylla* (1.62%), *Artemisia annua* (1.62%), *Cymbopogon martinii* (2.05% w/w), *C. winterianus* (2.90%), *Mentha piperita* (2.22%), *Ocimum gratissimum* (3.38%), *Piper mollicomum* (2.98% w/w) and *P. regnelli* (2.70%).

3.2. Anti-*Escherichia coli* activity

MIC results of the oils obtained from the plants tested are shown in Table 2. Among the 29 medicinal species investigated, essential oil from *Cymbopogon martinii* exhibited a broad inhibition spectrum, presenting strong activity (MIC between 100 and 900 μg/mL) against three enterotoxigenic, two enteropathogenic, three enteroinvasive and one shiga-toxin serotypes, while *C. winterianus* inhibited strongly two enterotoxigenic, one enteropathogenic, one enteroinvasive and two shiga-toxin producers serotypes. *Aloysia triphylla* also shows good potential to inhibit the growth of *Escherichia coli* with moderate to strong inhibition. Other essential oils showed antimicrobial properties, however with a more restrict action against the serotypes studied.

There is not an agreement on the acceptable inhibition level for plant materials when compared with standards. According to Aligiannis et al. (2001), who proposed a classification of plant materials based on MIC results (strong inhibitors: MIC up to 500 μg/mL; moderate inhibitors: MIC between 600 and 1500 μg/mL; weak inhibitors: MIC above 1600 μg/mL), the essential oil from *Cymbopogon martinii* can be considered a strong inhibitor of *Escherichia coli*, since it inhibited 10/13 serotypes tested with MIC values ≤ 500 μg/mL.

Table 2
Essential oil yield (%) and MIC results (μg/mL) from Brazilian plants tested against *Escherichia coli* serotypes

Medicinal plant	Oil yield (%w/w)	ETEC TR 441	ETEC 6/81H5J	ETEC 063	ETEC 5041-1	EPEC 0551	EPEC 0119	EPEC E234869	EPEC 0031-2	EIEC 1381-7	EIEC 0461-4	EIEC 240-1	STEC 2781-8	STEC 0157
Achyrocline satureioides (Lam.) DC.	0.44	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	900	1000	900	900
Allium schoenoprasum L.	0.72	[a]	[a]	[a]	1000	[a]	500	[a]	900	600	700	600	[a]	400
Aloysia triphylla (L'Hér.) Britton	1.60	600	1000	[a]	500	800	400	800	500	700	700	800	600	500
Anthemis nobilis L.	0.81	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	1000	[a]	800	1000	400
Artemisia annua L.	1.62	[a]	[a]	[a]	800	[a]	[a]	[a]	500	[a]	[a]	800	900	500
Artemísia camphorata L.	1.00	[a]	[a]	[a]	500	[a]	600	[a]	600	[a]	[a]	600	700	400
Baccharis trimera DC.	0.22	[a]	[a]	[a]	600	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]
Cordia verbenacea DC.	0.10	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]
Cymbopogon martinii (*Roxb.*) *W. Watson*	2.05	400	500	[a]	100	[a]	200	900	200	500	500	400	500	300
C. winterianus Jowitt	2.90	800	500	[a]	500	600	[a]	800	200	800	700	400	600	500
Cyperus articulatus L.	0.64	1000	[a]	[a]	[a]	[a]	700	[a]	800	[a]	[a]	1000	[a]	800
C. rotundus L.	0.40	1000	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]
Lippia alba (Mill.) N. E. Br. ex Britton & P. Wilson	1.28	1000	[a]	[a]	[a]	[a]	[a]	[a]	400	[a]	1000	500	800	400
M. piperita L.	2.22	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	800	800	1000	[a]	700
Mikania glomerata Spreng.	0.15	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]
M. laevigata Sch. Bip. ex Baker	0.08	[a]	[a]	[a]	500	[a]	[a]	[a]	500	900	[a]	600	600	300
Ocimum basilicum L.	0.49	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	200	500	200
O. gratissimum L.	3.38	900	[a]	[a]	1000	[a]	700	[a]	600	[a]	1000	700	800	600
Origanum × *applii* Boros	1.12	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	900	1000	900	1000	700
O. vulgare subsp. *virens* L.	0.68	[a]	[a]	700	[a]	[a]	600	[a]	[a]	[a]	[a]	500	[a]	400
Piper abutilodes Kunth	1.10	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	700
Piper aduncum L.	0.39	[a]	[a]	[a]	[a]	[a]	[a]	[a]	900	900	900	900	1000	500
P. marginatum Jacq.	1.24	[a]	[a]	[a]	[a]	[a]	[a]	[a]	900	[a]	[a]	[a]	[a]	700
P. molicomum Kunth	2.98	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	1000
P. regnellii (Miq.) C. DC.	2.70	[a]	[a]	[a]	[a]	[a]	[a]	[a]	300	[a]	[a]	[a]	[a]	1000
Stachys byzantina K. Koch	0.03	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	1000	[a]	900
Stachytarpheta cayennensis (*Rich.*) *Vahl*	0.04	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	900	[a]	900	700	900
Thymus vulgaris L.	2.66	[a]	[a]	[a]	[a]	[a]	[a]	[a]	1000	1000	1000	400	[a]	300
Urena lobata L.	0.20	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	[a]	1000	[a]	1000
Chloramphenicol	–	8.0	4.0	4.0	8.0	4.0	2.0	4.0	8.0	2.0	2.0	2.0	4.0	2.0

ETEC, enterotoxigenic *Escherichia coli*; EPEC, enteropathogenic *Escherichia coli*; EIEC, enteroinvasive *Escherichia coli*; STEC, shiga-toxin producer *Escherichia coli*.
[a] MIC > 1000 μg/mL.

Table 3
Identified compounds from the essential oil of *Cymbopogum martinii* (CM)

Compounds	RI[a]	RI[b]	*Cymbopogon martini* oil[c]
β-Myrcene	990	1134	0.12
Cis-β-ocimene	1034	1210	0.23
Trans-β-ocimene	1044	1226	0.98
Trans-geraniol	1254	1789	63.46
Geranyl acetate	1382	1750	28.83
Trans-caryophyllene	1414	1578	2.15
(*E*, *E*) farnesol	1720	–	1.57
Total			97.34

[a] RI = retention index, apolar column.
[b] RI = retention index, polar column.
[c] Results expressed as % of area.

Table 4
Antimicrobial activity of majority compounds from *Cymbopogon martinii* essential oil against two different *Escherichia coli* serotypes (MIC, μg/mL)

Compound	*Escherichia coli* ETEC 5041-1	*Escherichia coli* EPEC 0031-2
Geraniol	20	8.0
Geranyl acetate	500	400
trans-Caryophyllene	>1000	>1000
Cloramphenicol	8.0	8.0

3.3. Chemical composition of the Cymbopogon martinii essential oil

The essential oil from *Cymbopogon martinii*, which presented the best activity results, was subjected to GC–MS analyses (Table 3). The oil constituents were identified using the data sources available. The three major identified compounds were *trans*-geraniol (63.46%), geranyl acetate (28.83%) and in a lower concentration *trans*-caryophyllene (2.51%). Since they have been previously reported as antimicrobial (Scortichini and Rossi, 1991; Cruz et al., 1993; Pattnaik et al., 1997; Douglas et al., 2004; Mimica-Dukic et al., 2004) commercial samples of them were tested against one ETEC and one EPEC serotypes. Results showed (Table 4) that geraniol possessed the best antimicrobial activity (MICs = 20 and 8 μg/mL) being EPEC serotype the most susceptible species (MIC = 8 μg/mL). Geranyl acetate showed higher MICs (500 and 400 μg/mL, respectively) and *trans*-caryophyllene was inactive against the two strains tested. Considering that geraniol constitutes more than 60% of *Cymbopogon martinii* essential oil, results obtained here suggest that it could be the main responsible for the antibacterial effect of the whole essential oil.

4. Conclusions

In the present study, 17 out of 29 species screened for anti-*Escherichia coli* activity have been indicated in Brazilian folk medicine as antiseptic, anti-infective and/or anti-inflammatory. With the exception of the essential oils of *C. verbenaceae* and *Mikania glomerata*, all species presented strong to moderate activity against at least one *Escherichia coli* serotype. The essential oils from *A. tryphilla*, *Cymbopogon martinii* and *C. winterianus* were able to inhibit the most of serotypes tested, being *Cymbopogon martinii* the most active specie showing anti-*Escherichia coli* activity against 10 out of 13 serotypes. The significant antibacterial activities of *Cymbopogon martinii* oil and of geraniol, suggest that they could be useful for treating diarrhoea caused by enteropathogenic strains of *Escherichia coli*.

Acknowledgement

Research was supported by a grant from FAPESP (SP, Brazil).

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